Module 9: Inheritance Genetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain Mendel's two laws
2. Use Punnett squares to predict offspring traits
3. Solve problems with incomplete dominance, codominance, and blood types
4. Analyze pedigrees
5. Explain polygenic traits and how environment affects phenotype

The Big Picture

> Mendel discovered that traits are passed from parents to offspring in predictable patterns. By understanding these patterns, we can predict the probability of traits in offspring.

Key Terms

Basic Genetics Vocabulary

- **Gene** — A segment of DNA that codes for a trait
- **Allele** — A version of a gene (e.g., "B" for brown, "b" for blue)
- **Genotype** — The alleles an organism has (e.g., Bb)
- **Phenotype** — What the organism looks like (e.g., brown eyes)
- **Homozygous** — Two same alleles (AA or aa)
- **Heterozygous** — Two different alleles (Aa)
- **Dominant** — The allele that shows up in the heterozygote (masks the recessive)
- **Recessive** — The allele that is hidden unless homozygous (aa)
- **Carrier** — A person who is Aa for a recessive disorder (has it but doesn't show it)

Mendel's Laws

- **Law of Segregation** — Each parent gives one allele to each offspring (alleles separate during meiosis)
- **Law of Independent Assortment** — Genes on different chromosomes are inherited independently

Important Ratios to Know

Cross	Ratio
Monohybrid (Aa × Aa)	3 dominant : 1 recessive
Test cross (Aa × aa)	1 dominant : 1 recessive
Dihybrid (AaBb × AaBb)	9:3:3:1
Incomplete dominance (Rr × Rr)	1:2:1 (red : pink : white)

Beyond Simple Dominance

- **Incomplete Dominance** — Heterozygote is a blend (red + white = **pink**)
- **Codominance** — Both alleles fully show (Type A + Type B = **Type AB**)
- **Sex-linked (X-linked)** — Gene is on the X chromosome; males only have one X, so one recessive allele = affected

ABO Blood Types

Genotype	Blood Type
$I^A I^A$ or $I^A i$	Type A
$I^B I^B$ or $I^B i$	Type B
$I^A I^B$	Type AB
i i	Type O

Pedigree Clues

- **Autosomal recessive** — Can skip generations; two unaffected parents → affected child
- **Autosomal dominant** — Appears every generation; affected child must have an affected parent
- **X-linked recessive** — More males affected; never father-to-son; carrier mothers pass to sons

Polygenic Traits

- Controlled by **many genes** (not just one)
- Produce a **range** of phenotypes (bell curve), not just two categories
- Examples: height, skin color, weight
- **Environment also matters** — nutrition affects height, sunlight affects skin color

Study Tips

1. **Practice Punnett squares** — Do them until they're automatic
2. **Memorize the ratios** — 3:1, 9:3:3:1, 1:2:1
3. **Blood types are a favorite exam topic** — Know the I^A, I^B, i system
4. **For pedigrees, ask two questions** — Dominant or recessive? Autosomal or X-linked?
5. **Connect to Module 8** — Segregation = homologs separating in Meiosis I